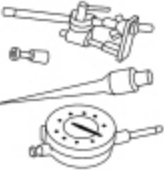
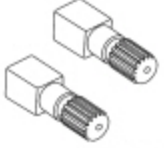
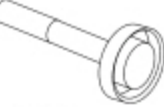


Differential Case and Ring Gear — Conventional

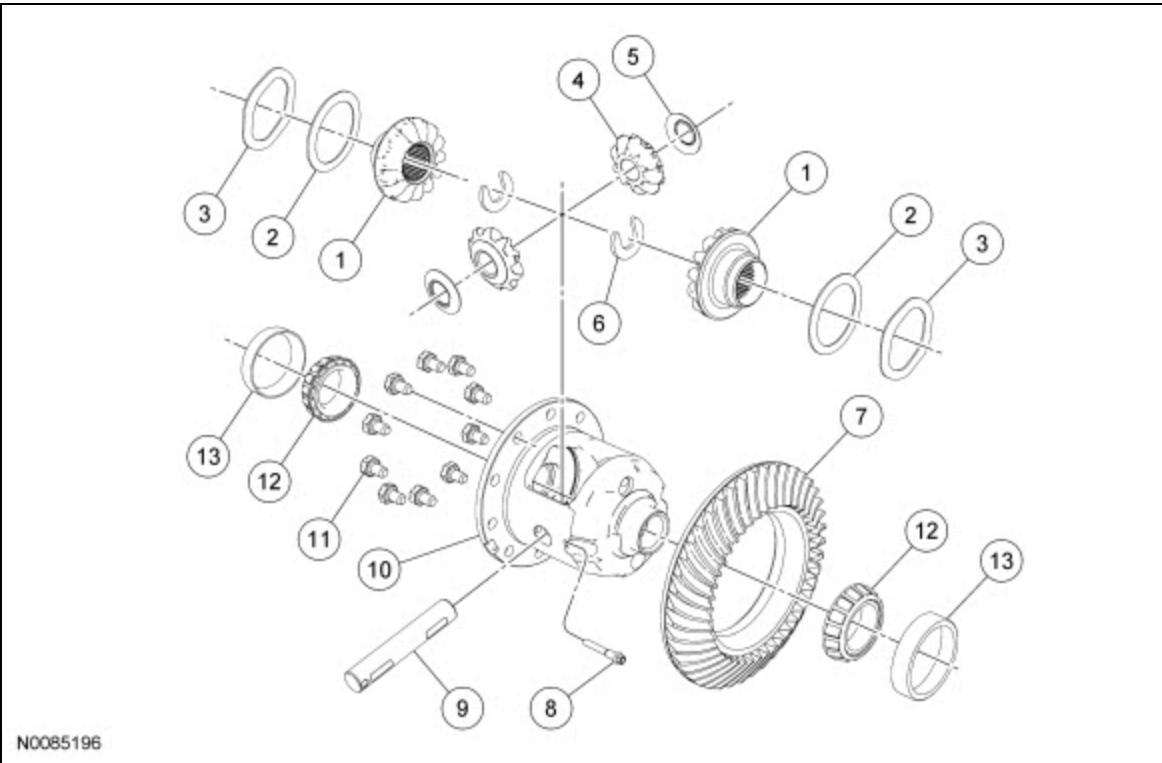


Special Tool(s)

 ST2026-A	2 Jaw Puller 205-D072 (D97L-4221-A) or equivalent
 ST1214-A	Dial Indicator Gauge With Holding Fixture 100-002 (TOOL-4201-C)
 ST1857-A	Gauge, Differential (Traction-Lock) 205-384 (T97T-4205-A)
 ST1749-A	Gauge, Differential (Traction-Lock) 205-386 (T97T-4205-D)
 ST2473-A	Installer, Differential Carrier Bearing 205-D044 (D81T-4221-A) or equivalent
 ST1543-A	Step Plate 205-D061 (D80L-630-8) or equivalent

Material

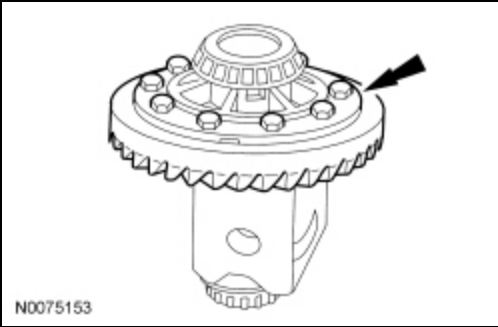
Item	Specification
Maximum Strength Retaining Compound Loctite® 638™	—
Motorcraft® SAE 75W-140 Synthetic Rear Axle Lubricant	WSL-M2C192-A and GL-5



Item	Part Number	Description
1	—	Differential side gears (part of 4215)
2	4228 4228	Differential side gear thrust washers
3	4214 4214	Differential side gear wave washers
4	4215 4215	Differential pinion gear
5	4230 4230	Differential pinion thrust washer (also part of 4215)
6	4N237 4N237	Axle shaft U-washer
7	4209 4209	Differential ring gear
8	4241 4241	Differential pinion shaft lock bolt (also part of 4215)
9	4211 4211	Differential pinion shaft
10	4204 4204	Differential case
11	4216 4216	Differential ring gear bolts (12 required)
12	4220 4220	Differential bearings
13	—	Differential bearing cups (part of 4220)

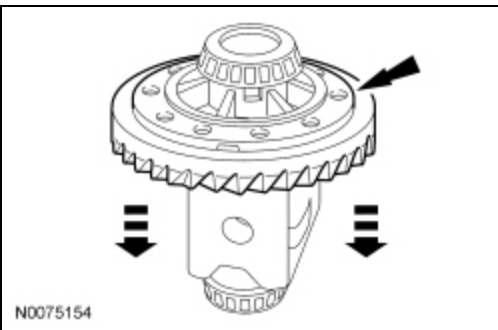
Disassembly

1. Remove the differential carrier. Tag the differential shims and bearing cups for location. For additional information, refer to Differential Carrier in this section.
2. Remove and discard the 12 differential ring gear bolts.



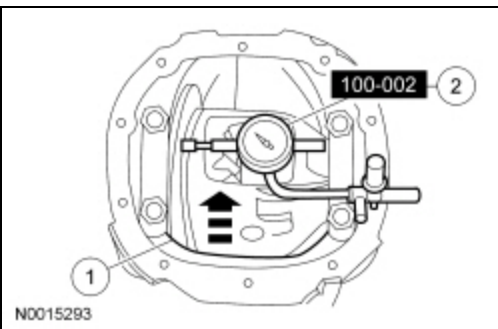
3. **NOTICE:** Care should be taken not to damage the differential ring gear bolt hole threads.

Insert a punch in the differential ring gear bolt holes and drive the differential ring gear off.

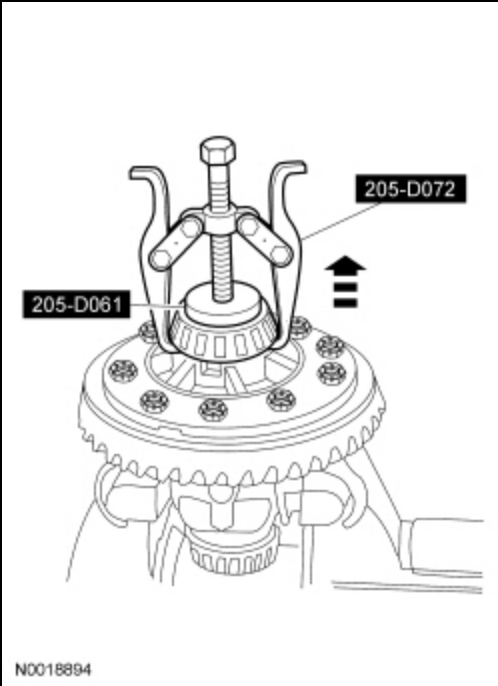


4. If checking differential case flange runout, install the bearing cups in their original location and install the differential assembly.

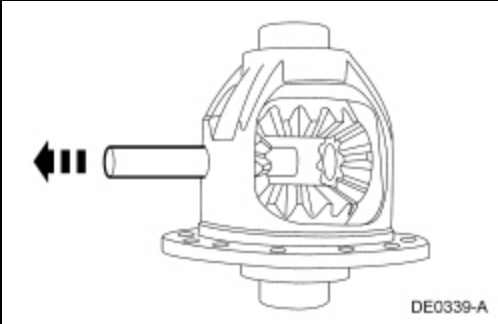
1. Install the bearing caps and differential shims.
 - Rotate the differential assembly to verify the bearings have seated.
2. Install the Dial Indicator Gauge With Holding Fixture and record the differential case runout.



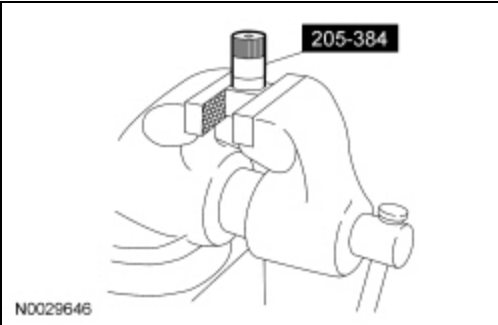
5. If necessary, remove the differential bearings with the 2 Jaw Puller and Step Plate.



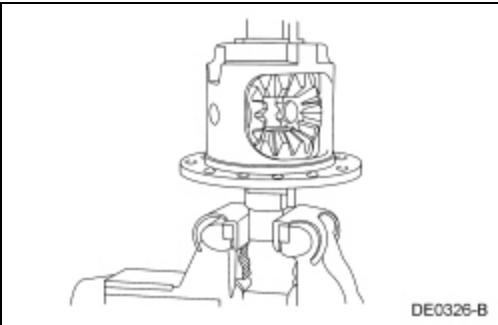
6. Remove the differential pinion shaft lock bolt and the differential pinion shaft.



7. Install the Differential Gauge in a suitable vise.

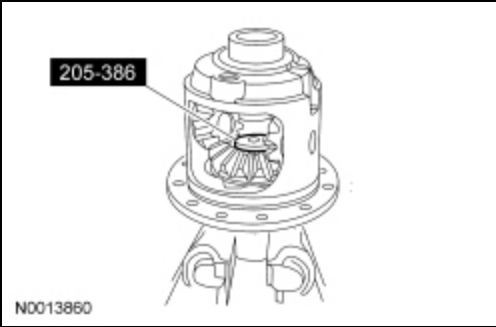


8. Install the differential case on the Differential Gauge.

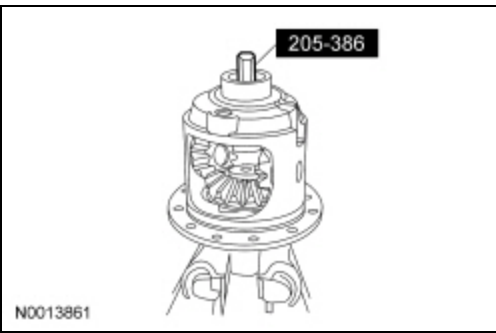


9. **NOTE:** Apply a small amount of grease to the centering hole of the Differential Gauge step plate.

Install the Differential Gauge step plate in the bottom differential side gear.

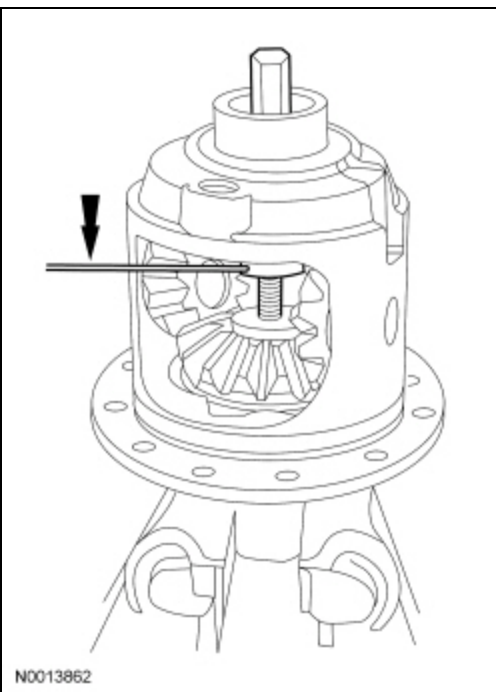


10. Install the Differential Gauge nut in the upper differential side gear. Hold the nut in position while installing the Differential Gauge hex screw. Tighten the hex-head screw until contact is made with the Step Plate.

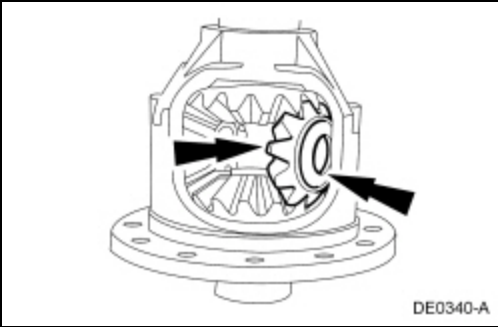


11. **NOTE:** *The dowel bar is used to keep the nut from turning when the forcing screw is tightened.*

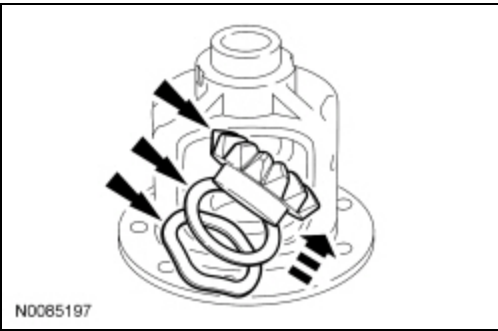
Insert a suitable dowel bar in the hole of the nut. Tighten the forcing screw to force the differential side gears away from the differential pinion gears.



12. Rotate and remove the differential pinion gears and differential pinion gear thrust washers.



13. Remove the differential side gears, the differential side gear thrust washers and the differential side gear wave washers.

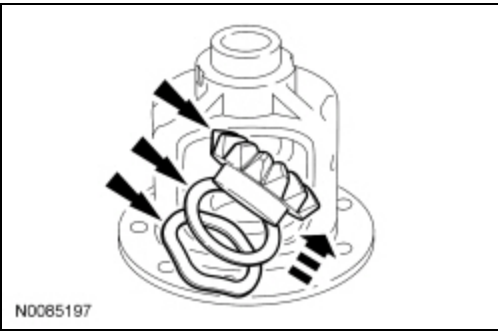


Assembly

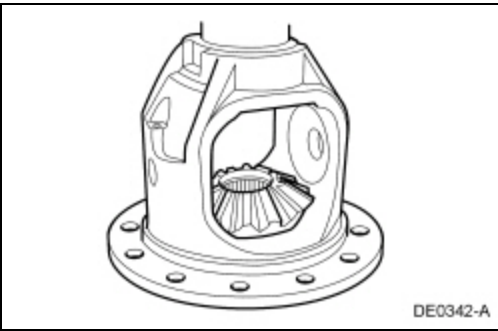
All vehicles

1. **NOTE:** Lubricate the differential side gear thrust washers, wave washers and the side gear journals with axle lubricant.

Position the differential side gear thrust washers and wave washers on the differential side gears.



2. Position the differential side gears.

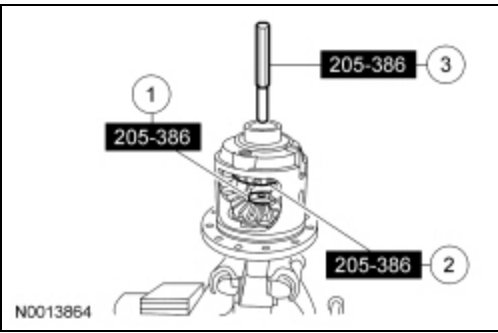


3. **NOTE:** Apply a small amount of grease to the step plate bore.

NOTE: If necessary, insert the dowel bar in the nut bore to keep the nut from turning as the hex screw is tightened.

Assemble the Differential Gauge to the differential case.

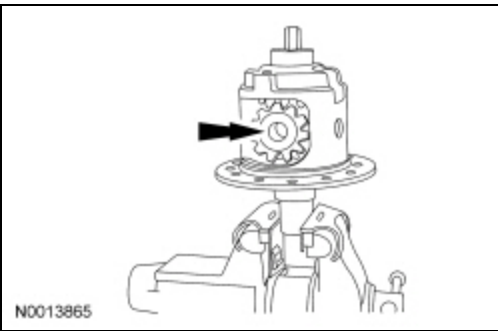
1. Position the Differential Gauge step plate in the bottom differential side gear.
2. Position the Differential Gauge hex screw plate in the top differential side gear and hold it in place.
3. Install the Differential Gauge hex screw and tighten it 2 turns after it contacts the bottom step plate.



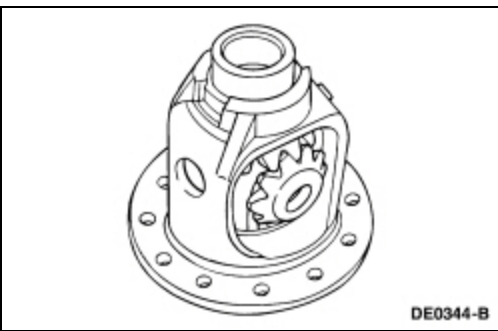
4. **NOTE:** Lubricate both sides of the differential pinion gear thrust washers with axle lubricant.

NOTE: Make sure the differential pinion gears are 180 degrees apart so they will align correctly with the pinion shaft bore.

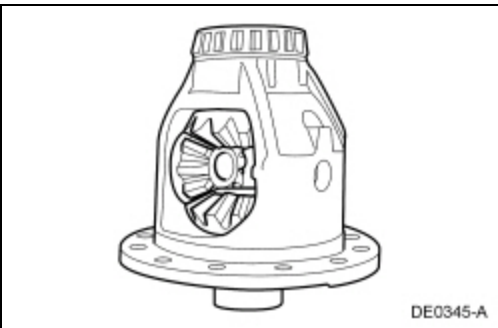
Position the differential pinion gears and differential pinion gear thrust washers in the window of the differential case so they mesh with the differential side gear teeth.



5. Engage the differential pinion gears opposite the differential side gears.

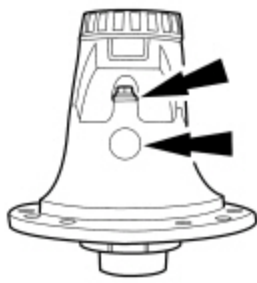


6. Rotate the differential pinion gears to align the differential pinion shaft bore.

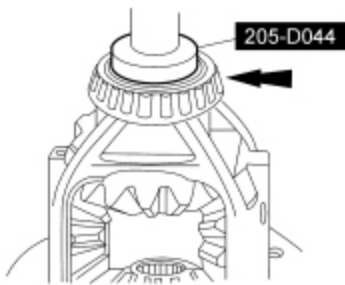


7. Remove the Differential Gauges after verifying the differential pinion shaft bore alignment.

8. Insert the differential pinion shaft, and install a new differential pinion shaft lock bolt finger-tight.



9. If removed, use the Differential Carrier Bearing Installer to install the differential bearings.



Electronic Locking Differential (ELD) equipped vehicles and all vehicles equipped with 3.55, 3.73, and 4.10 ratio gear sets

10. **NOTE:** *The differential flange and ring gear flange must be free of any old retaining compound material. Make sure both surfaces are clean and free of oil, dust and debris. Failure to clean the surfaces can result in ring gear runout concerns.*

Clean all traces of the old retaining compound material from the differential flange.

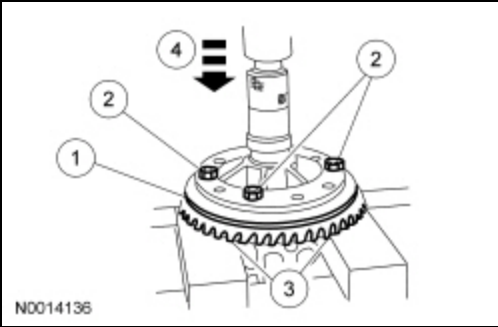
- Use solvent and Scotch-Brite® pads to remove.

11. Apply a one-eighth inch bead of maximum strength retaining compound on the rear face of the ring gear in the pattern shown.



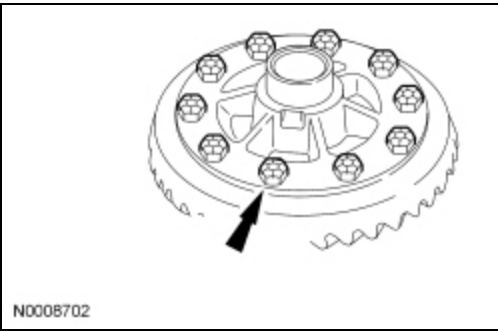
All vehicles

12. Install the ring gear.
1. Place the ring gear on the differential assembly.
 2. Hand-start 3 new ring gear bolts to align the holes in the ring gear and the differential case.
 3. Using a suitable press, position the ring gear.
 4. Press the ring gear into place.



13. Hand-start the remaining 9 new differential ring gear bolts. Tighten the 12 bolts in 2 stages.

- Stage 1: Tighten to 60 Nm (44 lb-ft).
- Stage 2: Tighten an additional 90 degrees.



14. Install the differential carrier. For additional information, refer to Differential Carrier in this section.